

Preregistration

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1 Quantitative Analyses

1.1 Interference

1.1.1 Between Subjects

2 linear models (daily updates and monthly updates)

$\text{interfere} \sim 1 + \text{gender_centered} + \text{race_centered} + \text{income_centered} + \text{geography_centered}$

Interpretation:

- Overall ratings of interference (are ratings, on average, different than neutral/0)
- Demographic effects (are ratings, on average, different across demographic groups)

1.1.2 Within Subjects

1 linear model (daily updates - monthly updates)

$\text{interfere_diff} \sim 1 + \text{gender_centered} + \text{race_centered} + \text{income_centered} + \text{geography_centered}$

Interpretation:

- Overall difference in ratings of interference by measure (are there relative differences in interference ratings by measure)
- Demographic effects (are differences in ratings of interference by measure, on average, different across demographic groups)

1.2 Dislike

5 linear models (daily updates, monthly updates, gps, SMS/phone logs, SMS content)

$\text{dislike} \sim 1 + \text{gender_centered} + \text{race_centered} + \text{income_centered} + \text{geography_centered}$

Interpretation:

- Overall ratings of dislike (are ratings, on average, different than neutral/0)
- Demographic effects (are ratings, on average, different across demographic groups)

1.2.1 Within Subjects

10 linear models (daily updates - monthly updates; daily updates - gps; daily updates - sms/phone logs; daily updates - sms content; monthly updates - gps; monthly updates - sms/phone logs;

monthly updates - sms content; gps - sms/phone logs; gps - sms content; sms/phone logs - sms content)

$\text{dislike_diff} \sim 1 + \text{gender_centered} + \text{race_centered} + \text{income_centered} + \text{geography_centered}$

Interpretation:

- Overall difference in ratings of dislike by measure (are there relative differences in dislike ratings by measure)

- Demographic effects (are differences in ratings of dislike by measure, on average, different across demographic groups)

2 Demo: Quantitative

Generate simulated data with the following variables:

Integer values ranging from -2 to 2 skewed negatively - daily_update_interfere - monthly_update_interfere - daily_update_dislike - monthly_update_dislike - gps_dislike - sms_phone_logs_dislike - sms_content_dislike

Centered demographic variables (-.5, .5) - gender - race - income - geography

```
set.seed(102030)

n <- 100

data <- tibble(subid = seq(1, 100, 1),
  daily_update_interfere = sample(c(-2, -1, 0, 1, 2),
    n,
    replace = TRUE,
    prob = c(.40, .30, .15, .10, .05)),
  monthly_update_interfere = sample(c(-2, -1, 0, 1, 2),
    n,
    replace = TRUE,
    prob = c(.40, .30, .15, .10, .05)),
  daily_update_dislike = sample(c(-2, -1, 0, 1, 2),
    n,
    replace = TRUE,
    prob = c(.45, .25, .15, .10, .05)),
  monthly_update_dislike = sample(c(-2, -1, 0, 1, 2),
    n,
    replace = TRUE,
    prob = c(.45, .25, .15, .10, .05)),
  gps_dislike = sample(c(-2, -1, 0, 1, 2),
    n,
    replace = TRUE,
    prob = c(.50, .25, .15, .07, .03)),
  sms_phone_logs_dislike = sample(c(-2, -1, 0, 1, 2),
    n,
    replace = TRUE,
    prob = c(.50, .25, .15, .07, .03)),
```

```

sms_content_dislike = sample(c(-2, -1, 0, 1, 2),
                             n,
                             replace = TRUE,
                             prob = c(.55, .20, .15, .07, .03)),
gender = sample(c(-.5, .5),
                n,
                replace = TRUE,
                prob = c(.50, .50)),
race = sample(c(-.5, .5),
              n,
              replace = TRUE,
              prob = c(.70, .30)),
income = sample(c(-.5, .5),
                n,
                replace = TRUE,
                prob = c(.50, .50)),
geography = sample(c(-.5, .5),
                   n,
                   replace = TRUE,
                   prob = c(.60, .40))) |>
glimpse()

```

Rows: 100

Columns: 12

```

$ subid          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14~
$ daily_update_interfere <dbl> 2, -2, 1, 1, -2, -1, -2, -1, -2, -2, -2, -
2, ~
$ monthly_update_interfere <dbl> 2, -1, -1, 2, -2, -2, 1, -2, 0, 0, 1, 0, -
2, ~
$ daily_update_dislike    <dbl> -1, -2, 1, 0, 0, -2, -2, -2, -2, -1, 2, -
1, 1~
$ monthly_update_dislike  <dbl> 0, 0, -2, -1, -1, -2, 0, -1, 0, -2, 0, -
1, 1,~
$ gps_dislike            <dbl> -2, -2, -2, 1, -2, -2, -1, -2, -1, -1, -
1, 1,~
$ sms_phone_logs_dislike  <dbl> 0, 0, -2, -2, -2, -2, -1, -2, -2, 0, -2, -
2, ~
$ sms_content_dislike     <dbl> 2, 1, -2, -2, 0, -1, 1, -2, -2, -1, -1, 1, 2,~
$ gender                <dbl> -0.5, -0.5, -0.5, 0.5, -0.5, -0.5, 0.5, 0.5, ~
$ race                  <dbl> -0.5, -0.5, 0.5, -0.5, 0.5, -0.5, -0.5, -
0.5,~
$ income                <dbl> 0.5, 0.5, -0.5, 0.5, -0.5, -0.5, 0.5, 0.5, 0.~

```

```
$ geography <dbl> -0.5, -0.5, -0.5, -0.5, -0.5, 0.5, 0.5, 0.5, ~
```

2.1 Between Subjects: Daily Update Interference

- interpret overall effect (ratings different than neutral; intercept), gender effect, race effect, income effect, and geography effect on interference rating.

```
summary(lm(daily_update_interfere ~ 1 + gender + race + income + geography,  
           data = data))
```

Call:

```
lm(formula = daily_update_interfere ~ 1 + gender + race + income +  
    geography, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.7935	-0.8958	-0.1961	0.5848	3.1032

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.84452	0.13325	-6.338	7.77e-09 ***
gender	-0.37794	0.22602	-1.672	0.0978 .
race	0.51874	0.26892	1.929	0.0567 .
income	-0.07859	0.22665	-0.347	0.7296
geography	-0.30073	0.22576	-1.332	0.1860

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.114 on 95 degrees of freedom

Multiple R-squared: 0.08419, Adjusted R-squared: 0.04563

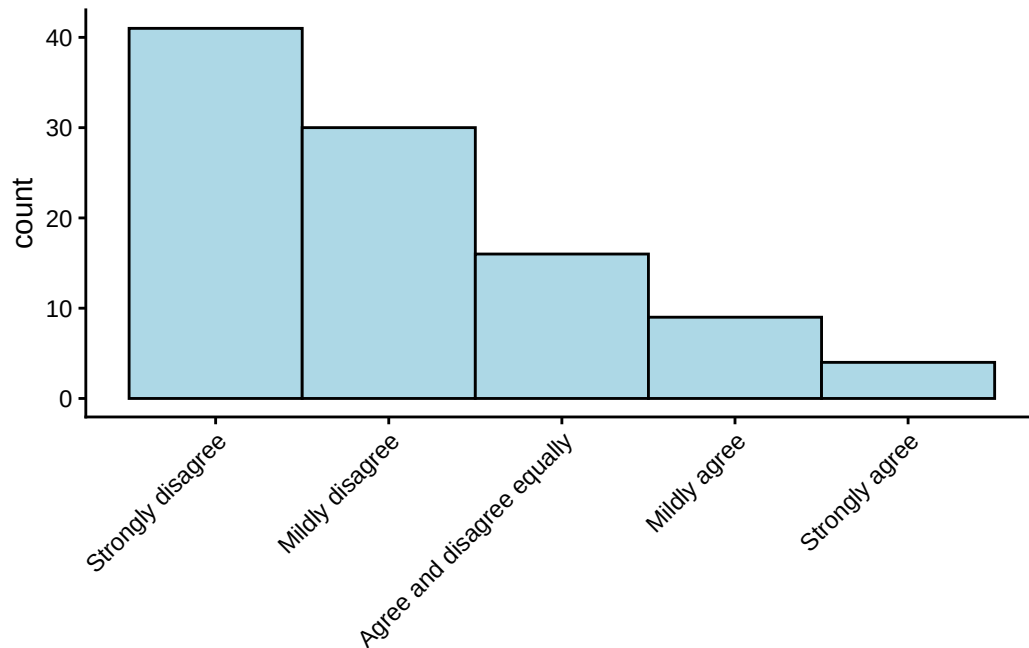
F-statistic: 2.183 on 4 and 95 DF, p-value: 0.07667

```
data |>  
  ggplot(aes(x = daily_update_interfere)) +  
  geom_histogram(bins = 5, color = "black", fill = "light blue") +  
  scale_x_continuous(NULL,  
                     breaks = c(-2, -1, 0, 1, 2),  
                     labels = c("Strongly disagree",  
                                "Mildly disagree",  
                                "Agree and disagree equally",
```

```

    "Mildly agree",
    "Strongly agree")) +
theme_classic() +
theme(axis.text.x = element_text(angle = 45,
    hjust = 1,
    vjust = 1))

```



2.2 Within Subjects: Daily vs Monthly Update Interference

```

data_within <- data |>
  mutate(interfere_diff = daily_update_interfere - monthly_update_interfere)

summary(lm(interfere_diff ~ 1 + gender + race + income + geography,
  data = data_within))

```

Call:

```

lm(formula = interfere_diff ~ 1 + gender + race + income + geography,
  data = data_within)

```

Residuals:

	Min	1Q	Median	3Q	Max
	-4.0918	-1.1491	0.2129	0.8846	3.8273

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.1323	0.1857	0.712	0.4780
gender	-0.4957	0.3150	-1.574	0.1188
race	0.3362	0.3748	0.897	0.3720
income	-0.6144	0.3159	-1.945	0.0547 .
geography	-0.5357	0.3146	-1.703	0.0919 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.553 on 95 degrees of freedom

Multiple R-squared: 0.1033, Adjusted R-squared: 0.06551

F-statistic: 2.735 on 4 and 95 DF, p-value: 0.03332

3 Qualitative Analyses

3.1 Overall Attitude

10 logistic regression models for each sensing method (daily update, monthly update, gps, phone/sms logs, sms content) x valence (positive, negative) combination.

attitude_positive ~ 1 + gender_centered + race_centered + income_centered + geography_centered

attitude_negative ~ 1 + gender_centered + race_centered + income_centered + geography_centered

Interpretation:

- Overall log-odds of positive/negative comment
- Demographic effects (are log-odds of positive/negative comments different by demographic group)

3.2 Theme Analyses

50 logistic regression models for each theme (benefits to self, benefits to others, privacy, research context, and feasibility) x sensing method (daily update, monthly update, gps, phone/sms logs, sms content) x valence (positive, neagative) combination.

theme_positive ~ 1 + gender_centered + race_centered + income_centered + geography_centered

theme_negative ~ 1 + gender_centered + race_centered + income_centered + geography_centered

Interpretation:

- Overall log-odds of positive/negative comment for theme
- Demographic effects (are log-odds of positive/negative comments for theme different by demographic group)

4 Demo: Qualitative

Generate simulated data with the following variables:

Dichotomous (0/1) values for coded themes - attitude_pos - attitude_neg - benefits_self_pos - benefits_self_neg - benefits_other_pos - benefits_other_neg - privacy_pos - privacy_neg - research_context_pos - research_context_neg - feasibility_pos - feasibility_neg

Centered demographic variables (-.5, .5) - gender - race - income - geography

```
set.seed(102030)

n <- 100

data <- tibble(subid = rep(1:n, 5),
               data = rep(c("daily_update",
                             "monthly_update",
                             "gps",
                             "sms_content",
                             "sms_phone_logs"),
                           each = n),
               attitude_pos = sample(c(1, 0),
                                     n*5,
                                     replace = TRUE,
                                     prob = c(.60, .40)),
               attitude_neg = sample(c(1, 0),
                                     n*5,
                                     replace = TRUE,
                                     prob = c(.60, .40)),
               benefits_self_pos = sample(c(1, 0),
                                           n*5,
```



```

                                replace = TRUE,
                                prob = c(.60, .40)),
benefits_self_neg = sample(c(1, 0),
                                n*5,
                                replace = TRUE,
                                prob = c(.20, .80)),
gender = sample(c(-.5, .5),
                                n*5,
                                replace = TRUE,
                                prob = c(.50, .50)),
race = sample(c(-.5, .5),
                                n*5,
                                replace = TRUE,
                                prob = c(.70, .30)),
income = sample(c(-.5, .5),
                                n*5,
                                replace = TRUE,
                                prob = c(.50, .50)),
geography = sample(c(-.5, .5),
                                n*5,
                                replace = TRUE,
                                prob = c(.60, .40))) |>

```

```
glimpse()
```

Rows: 500

Columns: 10

```

$ subid      <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
$ data       <chr> "daily_update", "daily_update", "daily_update", "dai~
$ attitude_pos <dbl> 0, 1, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0~
$ attitude_neg <dbl> 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1~
$ benefits_self_pos <dbl> 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 0, 1, 0, 0, 1, 1~
$ benefits_self_neg <dbl> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ gender      <dbl> -0.5, 0.5, -0.5, -0.5, -0.5, 0.5, 0.5, 0.5, 0.5, 0.5~
$ race        <dbl> -0.5, 0.5, 0.5, -0.5, -0.5, -0.5, -0.5, -0.5, -
0.5, ~
$ income      <dbl> -0.5, 0.5, 0.5, 0.5, -0.5, -0.5, 0.5, 0.5, -0.5, 0.5~
$ geography   <dbl> -0.5, 0.5, -0.5, 0.5, -0.5, -0.5, -0.5, 0.5, -
0.5, --

```

4.1 Overall Attitude Analysis: Positive and Negative for Daily Update

```
summary(lm(attitude_pos ~ 1 + gender + race + income + geography,  
           data = subset(data, data == "daily_update")))
```

Call:

```
lm(formula = attitude_pos ~ 1 + gender + race + income + geography,  
    data = subset(data, data == "daily_update"))
```

Residuals:

Min	1Q	Median	3Q	Max
-0.7595	-0.5090	0.2405	0.3600	0.6386

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.56046	0.05327	10.522	<2e-16 ***
gender	0.07568	0.09848	0.769	0.4441
race	-0.25053	0.10681	-2.346	0.0211 *
income	0.10378	0.09779	1.061	0.2912
geography	-0.11946	0.09753	-1.225	0.2237

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4795 on 95 degrees of freedom

Multiple R-squared: 0.07295, Adjusted R-squared: 0.03391

F-statistic: 1.869 on 4 and 95 DF, p-value: 0.1223

```
summary(lm(attitude_neg ~ 1 + gender + race + income + geography,  
           data = subset(data, data == "daily_update")))
```

Call:

```
lm(formula = attitude_neg ~ 1 + gender + race + income + geography,  
    data = subset(data, data == "daily_update"))
```

Residuals:

Min	1Q	Median	3Q	Max
-0.6983	-0.5283	0.3315	0.4502	0.5720

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.58076	0.05590	10.390	<2e-16 ***
gender	-0.06503	0.10334	-0.629	0.531
race	0.08346	0.11209	0.745	0.458
income	-0.12181	0.10262	-1.187	0.238
geography	0.03530	0.10235	0.345	0.731

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5032 on 95 degrees of freedom

Multiple R-squared: 0.02384, Adjusted R-squared: -0.01726

F-statistic: 0.5801 on 4 and 95 DF, p-value: 0.6777

4.2 Theme Analysis: Benfits to Self Positive and Negative for Daily Update

```
summary(lm(benefits_self_pos ~ 1 + gender + race + income + geography,  
           data = subset(data, data == "daily_update")))
```

Call:

```
lm(formula = benefits_self_pos ~ 1 + gender + race + income +  
    geography, data = subset(data, data == "daily_update"))
```

Residuals:

Min	1Q	Median	3Q	Max
-0.7157	-0.5189	0.2843	0.4376	0.6042

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.55573	0.05570	9.977	<2e-16 ***
gender	-0.08474	0.10298	-0.823	0.413
race	-0.05459	0.11170	-0.489	0.626
income	-0.11199	0.10226	-1.095	0.276
geography	0.06851	0.10199	0.672	0.503

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5014 on 95 degrees of freedom

Multiple R-squared: 0.03062, Adjusted R-squared: -0.01019

F-statistic: 0.7502 on 4 and 95 DF, p-value: 0.5603

```
summary(lm(benefits_self_neg ~ 1 + gender + race + income + geography,
           data = subset(data, data == "daily_update")))
```

Call:

```
lm(formula = benefits_self_neg ~ 1 + gender + race + income +
    geography, data = subset(data, data == "daily_update"))
```

Residuals:

Min	1Q	Median	3Q	Max
-0.36506	-0.22728	-0.16286	-0.08059	0.91941

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.22283	0.04557	4.890	4.1e-06 ***
gender	-0.08227	0.08424	-0.977	0.331
race	0.01493	0.09137	0.163	0.871
income	-0.06442	0.08365	-0.770	0.443
geography	0.12285	0.08343	1.472	0.144

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4102 on 95 degrees of freedom

Multiple R-squared: 0.03656, Adjusted R-squared: -0.004011

F-statistic: 0.9011 on 4 and 95 DF, p-value: 0.4666